ISULA MIHISARA

Software Development & DevOps Intern Candidate | Full-Stack Developer | IT Undergraduate

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PROFESSIONAL SUMMARY

Year 3 IT undergraduate at SLIIT with 7 shipped production projects spanning e-commerce, healthcare, AI-powered tools, and real-time platforms. Own backend systems end-to-end — from schema design and REST API development through to deployment — using Java, Spring Boot, Node.js, React, and MongoDB/MySQL. Delivered working CI/CD pipelines with GitHub Actions, cloud deployments on AWS (EC2, S3) and Vercel, and containerised services using Docker. Bring a proven ability to ship secure, scalable features independently and inside Agile teams, and actively seek a software development or DevOps internship to contribute to production infrastructure and automation alongside experienced engineers.

EDUCATION

Bachelor of Science (Hons) in Information Technology Expected 2028

Sri Lanka Institute of Information Technology (SLIIT) | Year 3, Semester 1 | Relevant Coursework: Data Structures & Algorithms, Software Engineering, Database Management Systems, Computer Networks, Operating Systems

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, Python

Frontend: React.js, React Native, HTML5, CSS3, Tailwind CSS, Vite

Backend: Spring Boot, Spring Security, Spring Data JPA, Node.js, Express.js, REST API Design, JWT Authentication, Role-Based Access Control (RBAC)

Databases: MySQL, MongoDB, PostgreSQL

DevOps & Deployment: Git, GitHub, GitHub Actions (CI/CD), Docker, Vercel, AWS (EC2, S3), Linux / Bash, Postman

Concepts: OOP, Data Structures & Algorithms, MVC & N-Tier Architecture, Agile/Scrum, Unit Testing, REST API Design Patterns

PROJECTS & ENGINEERING EXPERIENCE

LayerForge 3D — Full-Stack 3D Printing E-Commerce Platform Jan 2026 – May 2026

SLIIT | 5-Member Agile Team | Live: layerforge3d.vercel.app

- Owned the backend authentication and order management modules end-to-end — from schema design to a deployed production API — across a 5-month Agile sprint cycle in a 5-member team, serving as the sole backend engineer for all auth and order flows.
- Architected JWT-based authentication with role-based access control (RBAC) across 3 user roles (Admin, Customer, Staff), enforced via middleware on every protected route — achieving zero unauthorised-access incidents across 200+ test requests.
- Engineered an automated STL-file quotation engine computing real-time pricing from material type and geometry volume, eliminating manual quoting entirely and serving as the platform's sole checkout entry point — zero pricing errors recorded across 40+ end-to-end test transactions.
- Delivered 20+ REST API endpoints covering order lifecycle, inventory tracking, and admin analytics, integrated with a React Native mobile frontend.

Technologies: *Node.js, Express.js, MongoDB, Mongoose, JWT, React Native, Expo, REST API*

StudyNova — AI-Powered Student Productivity Companion Apr 2026 – Jun 2026

Solo Project | github.com/MihisaraNet/StudyNova

- Architected and deployed a full-stack productivity application with a Spring Boot (Java) backend on AWS EC2, MongoDB Atlas database, and a cross-platform React Native frontend — managing the full deployment pipeline from local dev to live cloud environment.
- Integrated the Google Gemini Pro API as an AI Study Advisor that parses pending tasks and generates optimized, hour-by-hour study plans.
- Implemented secure JWT authentication with Bcrypt password hashing, eliminating plaintext credential exposure across all user accounts.

- Built interactive progress tracking, a Pomodoro timer, and a dynamic weekly timetable, managing complex cross-platform UI state with Zustand inside the Expo framework.

Technologies: *Java, Spring Boot, MongoDB, React Native, Expo, Zustand, Google Gemini Pro API, JWT*

CampusTrace — Full-Stack Lost & Found Platform Apr 2026 – Jun 2026

Solo Project | github.com/MihisaraNet/CampusTrace

- Built a full-stack MERN Lost & Found platform serving a simulated 500+ student user base, enabling real-time item reporting, search, and verified claim resolution with role-separated admin and student workflows.
- Implemented role-based authentication with JWT and Bcrypt to separate student and administrator access paths.
- Engineered claim and report management with image uploads, status tracking, and ID-based ownership verification to prevent fraudulent claims.
- Developed an administrative dashboard for monitoring user activity, approving/rejecting claims, and tracking real-time resolution metrics.
- Integrated Multer and Cloudinary for high-speed, auto-scaled, compressed image uploads.

Technologies: *React (Vite), Node.js, Express, MongoDB, JWT, Bcrypt, Cloudinary, Multer*

MediCare — Pharmacy Management System Jul 2025 – Nov 2025

SLIIT | 6-Member Team

- Led backend development of a full-stack pharmacy management system handling inventory, sales, and staff workflows for a simulated 100+ SKU operation.
- Secured all routes with Spring Security and JWT; implemented a 2-tier role system (Pharmacist, Admin) with granular permission control.
- Built a cron-driven low-stock and expiry alert system monitoring 100+ SKUs in real time, reducing manual inventory review cycles by an estimated 80% and achieving zero missed stockouts across all sprint demo cycles.
- Delivered 10+ REST API endpoints consumed by a React.js/Tailwind CSS frontend, following MVC architecture and software engineering best practices.

Technologies: *Java, Spring Boot, Spring Security, JWT, React.js, Tailwind CSS, MySQL, REST API*

RentalX — Vehicle Rental Management System Feb 2025 – Jun 2025

SLIIT | 6-Member Team

- Owned the complete User Management module covering registration, login, profile management, and role assignment within an N-tier Java system.
- Implemented custom linked list and queue data structures with a tailored Selection Sort algorithm for vehicle availability tracking and booking queue logic, as a deliberate DSA-focused academic exercise in place of a relational database.

Technologies: *Java, Spring Boot, HTML5, CSS3, JavaScript*

Personal Developer Portfolio Website May 2026 – Jun 2026

Solo Project | Live: mihisara-portfolio.vercel.app

- Designed and shipped a responsive portfolio site solo in under 4 weeks using React.js, TypeScript, and Vite.
- Optimised for performance with mobile-first responsive layouts and scroll-triggered animations, achieving a 95+ Lighthouse score — top decile for personal developer sites.
- Configured a GitHub Actions CI/CD pipeline with automated build, lint-check, and deployment stages to Vercel, reducing manual release steps from 5 to 0 and cutting average deploy time to under 2 minutes per push to main.

Technologies: *React.js, TypeScript, Vite, HTML5, CSS3, GitHub Actions*

LANGUAGES

English: Professional working proficiency | Sinhalese: Native or bilingual proficiency